

NEELESH SARATHY

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EDUCATION

Purdue University <i>Bachelor's of Science, Computer Science and Data Science</i> <i>Certificate of Entrepreneurship and Innovation</i> GPA: 3.69 / 4.0	West Lafayette, IN May 2026
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PROFESSIONAL EXPERIENCE

Verma Lab Purdue University <i>Research Assistant - Data Science</i>	West Lafayette, IN May 2024 - Present
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Assisting biomedical researchers in data lifecycle management and machine learning.

- Constructing and maintaining data pipelines to power research projects and team workflows.

Eli Lilly - Sponsored research project.

- Developing predictive models for research applications.

Conducted Predictive Analysis for detecting diseases in animals using weather data.

- Analyzed complex weather pattern datasets to derive actionable insights.
- Developed a Random Forest model to forecast disease outcomes with 80% accuracy.

Implemented automatic monitoring of behaviors in farm animals.

- Built a model based on YOLOv8 for recognizing animal behaviors, achieving 95% accuracy.
- Facilitated elimination of man hours for farmers.

School of Civil and Construction Engineering Purdue University <i>Research Assistant - Data Science</i>	West Lafayette, IN August 2025 - Present
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Investigate the influence of state incentives on Electric Vehicle adoption.

- Evaluate and Benchmark Machine Learning methods for predictive accuracy.
- Produce interpretable insights to support forthcoming publication.

Atcharam Educational and Social Welfare Charitable Trust <i>Software Engineering Intern (Pro Bono)</i>	Tiruppur, India Remote August 2025 - Present
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Built and deployed a website from scratch for non-profit charitable trust using VueJS and Django.

- Maintain and enhance the site by implementing new features and updates as needed.
- Monitor website analytics and provide insights to support outreach and engagement.

College of Health and Human Sciences Purdue University <i>Research Assistant - Data Engineering</i>	West Lafayette, IN September 2023 - May 2024
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Analyzed social media sentiment on opinions regarding HIV vaccination.

- Scraped data from Reddit and parsed it from JSON format to a relational database.
- Implemented topic modeling to uncover subtopics within data.
- Reduced 7 man hours per week by automating data collection and parsing.

PROJECTS

Coffy - Embedded Database Engine [coffydb.org]	Plano, TX Remote May 2025 - Present
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- Created a unified local database engine with NoSQL, SQL, and graph databases.
- Devised lightweight ORM and query tools enabling rapid data prototyping.
- Demonstrated real world applications. [github.com/nsarathy/tetris-arcade-machine]

TravelTrack - Travel Log Application [github.com/nsarathy/TravelTrack]	West Lafayette, IN January 2025-May 2025
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- Developed a progressive web app for travel logging using Django and Vue.js.
- Utilized SQLite for data management.

SKILLS

Data Science	- Python, R, Pandas, Scikit-Learn, OpenCV
Web Development	- VueJS, Django, Java, JavaScript, HTML, CSS
Databases	- SQL, MongoDB, Neo4j
Systems	- C, C++, Linux, Bash Scripting

LEADERSHIP AND INVOLVEMENT

Purdue Boxing Club & Rhino's Gym <i>Captain</i>	Lafayette, IN August 2023 - Present
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- Teaching boxing fundamentals to newcomers during club practices.
- Organizing monthly club events.

PUBLICATIONS

List of Publications - Scholar.NeelSarathy.com